sensitivities. Also notice that the side of the order for the customer will be either +1 or −1 in the full iteration depending if the side was specified to be a buy order or a sell order respectively. The side of the order from all other market participants will be randomized (50% chance the trade was initiated from a buy order and 50% chance the trade was initiated from a sell order). The side of the order in our example only affects the market impact cost of the trade.

**Step 4.** Compute the average execution price of the customer's trade using the simulated price data above.

**Step 5.** Compute the customer trade cost as the difference between the average execution price and the starting price (adjusted for the side of the order).

**Step 6.** Repeat this experiment 22 times (one month of data) changing the customer's order size and side designations in each iteration.

**Step 7.** Plot the customer's trading cost on each day as a function of order size.

**Step 8.** Estimate the market impact sensitivity for one month of data and observe how close the estimated value is to the true value.

**Step 9.** Repeat this experiment several times to mimic several months of data. For each month estimate the market impact sensitivity.

Analysts are encouraged to perform these simulation exercises changing the variables above such as the temporary impact dissipation rate, market participant side parameter, total volume on the day, beta, volatility, etc. This will further highlight the difficulty in uncovering the true relationship between market impact and customer order size.

Figure 12.6 illustrates the above simulation exercise for one month (22 trading days) of data. The order sizes range from 0% to 25% ADV. But the relationship we uncovered from the data indicates that cost and size are negatively related which would suggest that larger order sizes have lower costs. The reason we have difficulty in uncovering a relationship between customer order and impact is, as mentioned previously, actual price movement is often dominated by stock alpha, market movement, buying and selling pressure from other market participants, and volatility.

Readers who carry out this simulation exercise are sure to have a difficult time estimating accurate market impact sensitivities using the customer order. In addition, this exercise also shows how unstable these parameters could be month to month (readers are encouraged to simulate several months worth of data and estimate parameters in each month). Notice that we used a very simple market impact model above and very basic assumptions and we